

UNIVERSITY COLLEGE LONDON

EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : **COMP0178**

ASSESSMENT PATTERN: **COMP0178A7PF 001**

MODULE NAME : **COMP0178 - Database Fundamentals**

LEVEL : **Postgraduate**

DATE: : **14/05/2024**

TIME : **14:30**

DURATION : **2 Hours**

This paper is suitable for candidates who attended classes for this module in the following academic year(s):

**Year
2023/24**

**EXAMINATION PAPER CANNOT BE REMOVED FROM THE EXAM HALL. PLACE
EXAM PAPER AND ALL COMPLETED SCRIPTS INSIDE THE EXAMINATION
ENVELOPE**

Hall instructions	N/A
Additional materials	N/A
Standard Calculators	Yes
Non-Standard Calculators	No

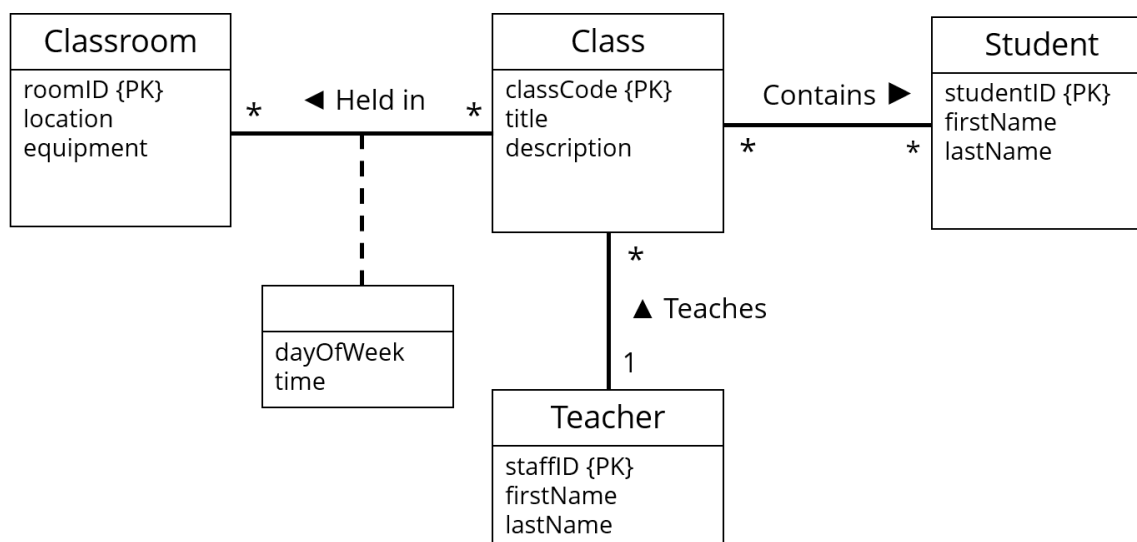
TURN OVER

Suitable for Cohort(s): 2023/24, 2022/23, 2021/22

There are FIVE questions in total. Answer ALL questions. Standard calculators are permitted.

A maximum of 100 marks is available. The marks available for each part of each question are indicated in square brackets [n].

- 1) [Database design] Consider the following Entity Relationship Diagram depicting a school system.



Convert the ER diagram into a relational database schema. Briefly explain (in 2-3 sentences) how you translated elements of the ER diagram and why you made those translations.

[Total for Question 1: 15 marks]

- 2) [Query processing] Consider the following database schema used to support the operations of a surgery.

Appointments (apptID, patientID, physicianID, date, time, room)

Patients (patientID, fName, lName, email, tel, preferredContact)

Physicians (physicianID, fName, lName, specialty)

[Question 2 continues over page]

[Question 2 continued]

- (a) Write an SQL query to answer a question that the surgery might have. An example of such a question would be the last name of all patients and physicians scheduled to use room 3A today. Your query should require all three tables.

[5 marks]

- (b) Translate the SQL query you wrote in the previous part to a relational algebra expression that accomplishes the same task.

[5 marks]

- (c) Translate the relational algebra expression you gave above into (1) a relational algebra tree that matches exactly the expression, and (2) a second relational algebra tree that is different from (not identical to) but equivalent in meaning to the first. For the relational algebra trees you've given, estimate the number of disk accesses that would be required under the following assumptions:

- No indexes exist.
- Tuples are read one at a time (1 tuple = 1 disk read).
- All tables fit entirely in memory.
- All intermediate calculations are stored to disk. However, the final result is returned directly to the user and not stored.
- The Appointments table contains 40,000 rows, Patients table contains 15,000 rows, and the Physicians tables contains 8 rows. There are roughly 20 appointments per physician per day.

If there is any unknown information needed to make your calculation (for example, a statistic like the number of rows resulting from a selection must be invented in order to complete a calculation), clearly state the assumptions you make.

[10 marks]

[Total for Question 2: 30 marks]

3) [Transactions]

- (a) Is the following interleaving conflict serializable or not? Explain your answer. In addition, if it is conflict serializable, what is the equivalent serial schedule?

T1: Read(x)
T2: Read(x)
T1: Write(x)
T3: Read(y)
T2: Write(y)
T3: Read(x)

[10 marks]

- (b) Under a timestamping-based concurrent transaction management scheme, what is the result of the following actions being taken? Does the following set of actions execute completely or does an issue arise at some point? Explain your answer.

T1: Read(A)
T2: Read(A)
T1: Write(A)
T2: Read(B)
T3: Read(A)
T2: Write(B)
T3: Read(B)

The numbers indicate the timestamp of each transaction.

[10 marks]

- (c) Describe two options for dealing with potential deadlock issues in a concurrency manager that uses locks.

[5 marks]

[Total for Question 3: 25 marks]

4) [Distributed databases]

- (a) Consider a distributed database supporting a library system. There are three main tables:

[Question 4 continues over page]

[Question 4 continued]

User (userId, email, name, etc.)

UserLoanHistory (userId, loanItemId, dateBorrowed, dateReturned, etc.)

LoanItems (loanItemId, author, title, publisher, etc.)

There are three sites (S1, S2, and S3), each storing roughly one-third of the User and UserLoanHistory tables based on userId. All three sites store a copy of LoanItem info.

Say that you want to do a count of how many times an item with loanItemId 1002839 has been borrowed. Describe two different strategies that could be used to carry out this distributed query, and which one would be more efficient, if there is any difference.

[10 marks]

- (b) In the 2-phase commit (2PC) protocol, what should the request coordinator do if it recovers from a failure and sees that the last entry that has been recorded in its log for the current transaction is “Commit Decision”? Briefly explain your answer (1-2 sentences).

[5 marks]

[Total for Question 4: 15 marks]

5) [Non-relational databases/general]

- (a) What is one strength and one weakness that relational databases have compared to semi-structured databases?

[10 marks]

- (b) Give two examples of ways that hashing can be used in databases.

[5 marks]

[Total for Question 5: 15 marks]

END OF PAPER